GENERAL SCIENCE: CHEMISTRY FUNDAMENTALS

ATOM AND PERIODIC IDENTITY → BOND AND FORCE LADDER → STOICHIOMETRIC ACCOUNTING RAIL → STATE AND SOLUTION MATRIX → ACID BASE BUFFER CONTROL → ELECTROCHEMICAL TWO-CELL MAP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

• PRIMARY CORE FLOW • SUBORDINATE EXTRA • APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 ATOM AND PERIODIC IDENTITY MAP

ATOM AND PERIODIC IDENTITY MAP | PROTONS + NEUTRONS | ATOMIC NUMBER | ELEMENT IDENTITY | MASS NUMBER | DIFFERENT COMPARISON AXES | PERIODIC TABLE | ATOMIC NUMBER + RECURRING PROPERTIES

NUCLEUS → protons + neutrons

ATOMIC NUMBER → protons → element identity → MASS NUMBER → protons + neutrons → ISOTOPE / ISOBAR / ISOTONE → different comparison axes

CONCEPT / ARCHITECTURE
- NUCLEUS → protons + neutrons
- ATOMIC NUMBER → protons → element identity

MECHANISM / APPLICATION
- MASS NUMBER → protons + neutrons
- ISOTOPE / ISOBAR / ISOTONE → different comparison axes

READINESS / STATUS LIMIT
- PERIODIC TABLE → atomic number + recurring properties
- NUCLEUS → protons + neutrons

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PERIODIC TABLE → atomic number + recurring properties.

01

STAGE 01 BOND AND FORCE LADDER

BOND AND FORCE LADDER | ELECTRON TRANSFER + ION ATTRACTION | SHARED ELECTRON PAIR | DELOCALISED ELECTRONS | HYDROGEN BOND | VAN DER WAALS | WEAKER INTERPARTICLE FORCES | DO NOT COLLAPSE BOND AND FORCE CATEGORIES

IONIC → electron transfer + ion attraction

COVALENT → shared electron pair → METALLIC → delocalised electrons → HYDROGEN BOND / VAN DER WAALS → weaker interparticle forces

STARTING CONDITION / INPUT
- IONIC → electron transfer + ion attraction
- COVALENT → shared electron pair

SCIENTIFIC OR ENGINEERING MECHANISM
- METALLIC → delocalised electrons
- HYDROGEN BOND / VAN DER WAALS → weaker interparticle forces

OUTPUT / FAILURE BOUNDARY
- TRAP → do not collapse bond and force categories
- IONIC → electron transfer + ion attraction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → do not collapse bond and force categories.

02

STAGE 02 STOICHIOMETRIC ACCOUNTING RAIL

STOICHIOMETRIC ACCOUNTING RAIL | WRITE SPECIES | BALANCE EQUATION | CONVERT SUPPLIED QUANTITIES | COMPARE STOICHIOMETRIC REQUIREMENT | IDENTIFY LIMITING REAGENT | THEORETICAL PRODUCT | YIELD, ONLY WITH SUPPLIED DATA

WRITE SPECIES → BALANCE EQUATION → CONVERT SUPPLIED QUANTITIES → MOLES → COMPARE STOICHIOMETRIC REQUIREMENT → IDENTIFY LIMITING REAGENT → THEORETICAL PRODUCT → ACTUAL / THEORETICAL → YIELD, only with supplied data

STARTING CONDITION / INPUT
- WRITE SPECIES → BALANCE EQUATION
- CONVERT SUPPLIED QUANTITIES → MOLES

SCIENTIFIC OR ENGINEERING MECHANISM
- COMPARE STOICHIOMETRIC REQUIREMENT
- IDENTIFY LIMITING REAGENT → THEORETICAL PRODUCT

OUTPUT / FAILURE BOUNDARY
- ACTUAL / THEORETICAL → YIELD, only with supplied data
- WRITE SPECIES → BALANCE EQUATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ACTUAL / THEORETICAL → YIELD, only with supplied data.

03

STAGE 03 STATE AND SOLUTION MATRIX

STATE AND SOLUTION MATRIX | ARRANGEMENT + MOBILITY + SEPARATION | PHASE CHANGE | PHYSICAL IDENTITY RETAINED | MOLECULAR/IONIC DISPERSION | AMOUNT ON A STATED BASIS | EQUILIBRIUM LIMIT UNDER STATED CONDITIONS

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
SOLID / LIQUID / GAS → arrangement + mobility + separation	SOLUTION → molecular/ionic dispersion	SOLUBILITY → equilibrium limit under stated conditions
PHASE CHANGE → physical identity retained	CONCENTRATION → amount on a stated basis	SOLID / LIQUID / GAS → arrangement + mobility + separation

- WRITE SPECIES → BALANCE EQUATION
- CONVERT SUPPLIED QUANTITIES → MOLES

- COMPARE STOICHIOMETRIC REQUIREMENT
- IDENTIFY LIMITING REAGENT → THEORETICAL PRODUCT

- ACTUAL / THEORETICAL → YIELD, only with supplied data
- WRITE SPECIES → BALANCE EQUATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ACTUAL / THEORETICAL → YIELD, only with supplied data.

03

STAGE 03

STATE AND SOLUTION MATRIX

STATE AND SOLUTION MATRIX · ARRANGEMENT + MOBILITY + SEPARATION · PHASE CHANGE · PHYSICAL IDENTITY RETAINED · MOLECULAR/IONIC DISPERSION · AMOUNT ON A STATED BASIS · EQUILIBRIUM LIMIT UNDER STATED CONDITIONS

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
SOLID / LIQUID / GAS → arrangement + mobility + separation	SOLUTION → molecular/ionic dispersion	SOLUBILITY → equilibrium limit under stated conditions
PHASE CHANGE → physical identity retained	CONCENTRATION → amount on a stated basis	SOLID / LIQUID / GAS → arrangement + mobility + separation

SYSTEM / DEVICE / INSTITUTION
- SOLID / LIQUID / GAS → arrangement + mobility + separation
- PHASE CHANGE → physical identity retained

DECISIVE TECHNICAL DISTINCTION
- SOLUTION → molecular/ionic dispersion
- CONCENTRATION → amount on a stated basis

STATUS / UNIT / EVIDENCE LIMIT
- SOLUBILITY → equilibrium limit under stated conditions
- SOLID / LIQUID / GAS → arrangement + mobility + separation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOLID / LIQUID / GAS → arrangement + mobility + separation.

04

STAGE 04

ACID BASE BUFFER CONTROL LOOP

ACID BASE BUFFER CONTROL LOOP · ACID DONATES H+ <→ BASE ACCEPTS H PH · LOGARITHMIC ACTIVITY MEASURE · ACIDIC/BASIC CHARACTER REDUCED · CONSUMES LIMITED ADDED ACID/BASE · SALT SOLUTION · NEUTRAL DEPENDS ON IONS

CONCEPT / ARCHITECTURE
- ACID DONATES H+ <→ BASE ACCEPTS H pH → logarithmic activity measure
- NEUTRALISATION → acidic/basic character reduced

MECHANISM / APPLICATION
- BUFFER → consumes limited added acid/base
- SALT SOLUTION → acidic / basic / neutral depends on ions

READINESS / STATUS LIMIT
- ACID DONATES H+ <→ BASE ACCEPTS H pH → logarithmic activity measure
- NEUTRALISATION → acidic/basic character reduced

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SALT SOLUTION → acidic / basic / neutral depends on ions.

05

STAGE 05

ELECTROCHEMICAL TWO-CELL MAP

ELECTROCHEMICAL TWO-CELL MAP · GALVANIC CELL · CHEMICAL ENERGY TO ELECTRICAL ENERGY · ELECTROLYTIC CELL · ELECTRICAL ENERGY DRIVES CHEMISTRY · APPLIED ROUTES · SIGN TRAP · CELL TYPE CHANGES SIGNS, NOT REACTION LOCATIONS

OXIDATION → anode

REDUCTION → cathode → GALVANIC CELL → chemical energy to electrical energy → ELECTROLYTIC CELL → electrical energy drives chemistry

STARTING CONDITION / INPUT
- OXIDATION → anode
- REDUCTION → cathode

SCIENTIFIC OR ENGINEERING MECHANISM
- GALVANIC CELL → chemical energy to electrical energy
- ELECTROLYTIC CELL → electrical energy drives chemistry

OUTPUT / FAILURE BOUNDARY
- CORROSION / PLATING / EXTRACTION → applied routes
- SIGN TRAP → cell type changes signs, not reaction locations

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIGN TRAP → cell type changes signs, not reaction locations.

06

STAGE 06

KINETICS VERSUS EQUILIBRIUM

KINETICS VERSUS EQUILIBRIUM · RATE AXIS · COLLISIONS + ACTIVATION ENERGY · ALTERNATIVE LOWER-ENERGY PATHWAY · EQUILIBRIUM AXIS · FORWARD RATE EQUALS REVERSE RATE · FASTER APPROACH IN BOTH DIRECTIONS · NO SHIFT

RATE AXIS → collisions + activation energy

CATALYST → alternative lower-energy pathway → EQUILIBRIUM AXIS → forward rate equals reverse rate → CATALYST → faster approach in both directions

CONCEPT / ARCHITECTURE
- RATE AXIS → collisions + activation energy
- CATALYST → alternative lower-energy pathway

MECHANISM / APPLICATION
- EQUILIBRIUM AXIS → forward rate equals reverse rate
- CATALYST → faster approach in both directions

READINESS / STATUS LIMIT
- NO SHIFT → equilibrium position unchanged
- RATE AXIS → collisions + activation energy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EQUILIBRIUM AXIS → forward rate equals reverse rate.

06

STAGE 06

KINETICS VERSUS EQUILIBRIUM

KINETICS VERSUS EQUILIBRIUM | RATE AXIS | COLLISIONS + ACTIVATION ENERGY | ALTERNATIVE LOWER-ENERGY PATHWAY | EQUILIBRIUM AXIS | FORWARD RATE EQUALS REVERSE RATE | FASTER APPROACH IN BOTH DIRECTIONS | NO SHIFT

RATE AXIS → collisions + activation energy

CATALYST → alternative lower-energy pathway → EQUILIBRIUM AXIS → forward rate equals reverse rate → CATALYST → faster approach in both directions

CONCEPT / ARCHITECTURE

- RATE AXIS → collisions + activation energy
- CATALYST → alternative lower-energy pathway

MECHANISM / APPLICATION

- EQUILIBRIUM AXIS → forward rate equals reverse rate
- CATALYST → faster approach in both directions

READINESS / STATUS LIMIT

- NO SHIFT → equilibrium position unchanged
- RATE AXIS → collisions + activation energy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EQUILIBRIUM AXIS → forward rate equals reverse rate.

07

STAGE 07

ORGANIC AND POLYMER FAMILY TREE

ORGANIC AND POLYMER FAMILY TREE | FUNCTIONAL GROUPS | ADDITION POLYMER | NO SMALL-MOLECULE ELIMINATION | CONDENSATION POLYMER | SMALL MOLECULE ELIMINATED | THERMOPLASTIC <→ THERMOSET | REHEATING BOUNDARY

FUNCTIONAL GROUPS → alcohol

ADDITION POLYMER → no small-molecule elimination → CONDENSATION POLYMER → small molecule eliminated → THERMOPLASTIC <→ THERMOSET → reheating boundary

CONCEPT / ARCHITECTURE

- FUNCTIONAL GROUPS → alcohol
- ADDITION POLYMER → no small-molecule elimination

MECHANISM / APPLICATION

- CONDENSATION POLYMER → small molecule eliminated
- THERMOPLASTIC <→ THERMOSET → reheating boundary

READINESS / STATUS LIMIT

- HYDROGEL → cross-linked hydrophilic network
- FUNCTIONAL GROUPS → alcohol

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FUNCTIONAL GROUPS → alcohol | aldehyde | ketone | acid | amine | ester.

08

STAGE 08

MATERIALS STRUCTURE-PROPERTY WHEEL

MATERIALS STRUCTURE-PROPERTY WHEEL | TETRAHEDRAL NETWORK | LAYERED STRUCTURE | DISTINCT CARBON ARCHITECTURES | COMPOSITION + PROCESSING ALTER PROPERTIES | SAME ELEMENT CAN YIELD DIFFERENT MATERIAL BEHAVIOUR

CONCEPT / ARCHITECTURE

- DIAMOND → tetrahedral network
- GRAPHITE → layered structure

MECHANISM / APPLICATION

- GRAPHENE / FULLERENE / NANOTUBE → distinct carbon architectures
- ALLOY → composition + processing alter properties

READINESS / STATUS LIMIT

- RULE → same element can yield different material behaviour
- DIAMOND → tetrahedral network

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → same element can yield different material behaviour.

09

STAGE 09

FUEL CONVERSION FAULT TREE

FUEL CONVERSION FAULT TREE | OXIDATION + USEFUL HEAT + POSSIBLE POLLUTANTS | INCOMPLETE COMBUSTION | DIFFERENT PRODUCT BURDEN | CONTROLLED OXYGEN/STEAM CONVERSION | CARBON MONOXIDE + HYDROGEN DOMINANT ROUTE | OUTCOME FIREWALL | PRODUCT IS NOT EFFICIENCY OR CLIMATE PROOF

COMBUSTION → oxidation + useful heat + possible pollutants

INCOMPLETE COMBUSTION → different product burden → GASIFICATION → controlled oxygen/steam conversion → SYNGAS → carbon monoxide + hydrogen dominant route

CONCEPT / ARCHITECTURE

- COMBUSTION → oxidation + useful heat + possible pollutants
- INCOMPLETE COMBUSTION → different product burden

MECHANISM / APPLICATION

- GASIFICATION → controlled oxygen/steam conversion
- SYNGAS → carbon monoxide + hydrogen dominant route

READINESS / STATUS LIMIT

- OUTCOME FIREWALL → product is not efficiency or climate proof
- MUST REMEMBER: Chemistry fundamentals link atomic structure, mole/stoichiometry, bonding,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME FIREWALL → product is not efficiency or climate proof.

10

STAGE 10

ENVIRONMENTAL COMPARTMENT MAP

COMBUSTION → oxidation + useful heat + possible pollutants

INCOMPLETE COMBUSTION → different product burden → GASIFICATION → controlled oxygen/steam conversion → SYNGAS → carbon monoxide + hydrogen dominant route

CONCEPT / ARCHITECTURE
- COMBUSTION → oxidation + useful heat + possible pollutants
- INCOMPLETE COMBUSTION → different product burden

MECHANISM / APPLICATION
- GASIFICATION → controlled oxygen/steam conversion
- SYNGAS → carbon monoxide + hydrogen dominant route

READINESS / STATUS LIMIT
- OUTCOME FIREWALL → product is not efficiency or climate proof
- MUST REMEMBER: Chemistry fundamentals link atomic structure, mole/stoichiometry, bonding,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME FIREWALL → product is not efficiency or climate proof.

10 — STAGE 10 ENVIRONMENTAL COMPARTMENT MAP

ENVIRONMENTAL COMPARTMENT MAP | OZONE PROTECTIVE ROLE | OZONE POLLUTANT ROLE | NUTRIENTS, ACIDS, SALTS, PERSISTENT CHEMICALS | CONCENTRATION + ROUTE + DURATION + CONDITIONS | NOT SAFETY, HAZARD OR REGULATION EVIDENCE | CLOSE DISTINCTION | ATOM IS NOT MOLECULE, MOLE IS NOT MASS, IONIC

STRATOSPHERE → ozone protective role

TROPOSPHERE → ozone pollutant role → WATER / SOIL → nutrients, acids, salts, persistent chemicals → EXPOSURE → concentration + route + duration + conditions

CONCEPT / ARCHITECTURE
- STRATOSPHERE → ozone protective role
- TROPOSPHERE → ozone pollutant role

MECHANISM / APPLICATION
- WATER / SOIL → nutrients, acids, salts, persistent chemicals
- EXPOSURE → concentration + route + duration + conditions

READINESS / STATUS LIMIT
- LABEL → not safety, hazard or regulation evidence
- CLOSE DISTINCTION: Atom is not molecule, mole is not mass, ionic is not covalent,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Atom is not molecule, mole is not mass, ionic is not covalent,.

11 — STAGE 11 PYQ AND POLICY EVIDENCE LADDER

PYQ AND POLICY EVIDENCE LADDER | WATER POLARITY | MANY IONIC/POLAR SOLUTES, NOT EVERY SUBSTANCE | PRESENCE DIFFERS FROM EXPOSURE AND REGULATION | MATERIAL CLASS DIFFERS FROM UNIVERSAL PERFORMANCE | NOT DEPLOYMENT, OUTCOME, HAZARD OR LAW | EVIDENCE LIMIT | BALANCE SPECIES, CHARGE AND UNITS; STATE

WATER POLARITY → many ionic/polar solutes, not every substance → BPA / TRICLOSAN → presence differs from exposure and regulation → HYDROGEL → material class differs from universal performance → PRINCIPLE → PROCESS → APPLICATION → INSTITUTION → PAGE / DOCUMENT → not deployment, outcome, hazard or law

STARTING CONDITION / INPUT
- WATER POLARITY → many ionic/polar solutes, not every substance
- BPA / TRICLOSAN → presence differs from exposure and regulation

SCIENTIFIC OR ENGINEERING MECHANISM
- HYDROGEL → material class differs from universal performance
- PRINCIPLE → PROCESS → APPLICATION → INSTITUTION

OUTPUT / FAILURE BOUNDARY
- PAGE / DOCUMENT → not deployment, outcome, hazard or law
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Balance species, charge and units; state...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Balance species, charge and units; state...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Balance species, charge and units; state.

E — EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | CHEMISTRY, CATALYSIS, MATERIALS AND MOLECULAR RESEARCH. | ELECTROCHEMISTRY, CORROSION, BATTERIES AND RELATED TRANSLATIONAL TECHNOLOGIES. | METALLURGICAL ECOSYSTEM | EXTRACTION, ALLOY BEHAVIOR AND INDUSTRIAL MATERIALS. | NATIONAL GREEN HYDROGEN MISSION | LINKS ELECTROCHEMISTRY, CATALYSTS, STORAGE AND STANDARDS TO DECARBONIZATION STRATEGY. | UNEP PLASTICS-TREATY PROCESS

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH
- CSIR-NCL: chemistry, catalysis, materials and molecular research.
- CSIR-CECRI: electrochemistry, corrosion, batteries and related translational technologies.
- CSIR-NML / metallurgical ecosystem: extraction, alloy behavior and industrial materials.
- National Green Hydrogen Mission: links electrochemistry, catalysts, storage and standards to decarbonization strategy.

READINESS, UNIT OR STATUS LIMIT
- UNEP plastics-treaty process: international governance response to polymer lifecycle issues.
- Fertilizer chemistry is central to food security, soil management and subsidy debates.
- Electrochemistry matters for batteries, plating, corrosion control and hydrogen pathways.
- Polymer chemistry shapes packaging, healthcare disposables, pipes, textiles and waste burdens.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE
- Metallurgy underlies infrastructure, railways, defence, power systems and manufacturing competitiveness.
- Biomolecule chemistry supports diagnostics, therapeutics and industrial biotechnology.
- CAUTION: Limitation 1: chemical utility and environmental externality often move together
- — the same polymer that lowers cost may raise waste-management burden.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels